

Spandana Chereddy

PhD Researcher | Machine Learning & AI | Scientific AI | Computational Biology

✉ cheredds@uoguelph.ca  [Google Scholar](#)  [LinkedIn](#)

SUMMARY

PhD researcher in Computer Science specializing in machine learning, deep learning, scientific AI, and representation learning, with research spanning foundation models, multimodal learning, and scalable ML systems. Recipient of the **NSERC Canada Graduate Research Scholarship – Doctoral (CGRS-D)**, **PGS**, **Ontario Graduate Scholarship (OGS)**, **Vector Scholarship in Artificial Intelligence** and **DAAD Research Fellowship**. Research and engineering experience across the University of Guelph, Forschungszentrum Jülich, Microsoft, and the Centre for Biodiversity Genomics, with multiple peer-reviewed publications and international research presentations. Experienced in developing scalable ML models, research pipelines, and production software using Python, PyTorch, cloud technologies, and modern ML infrastructure.

Research Interests: Foundation Models, Multimodal Learning, Representation Learning, Scientific AI, Generative AI, Computational Biology

EDUCATION

University of Guelph

- Ph.D. in Computer Science

Guelph, Canada

Sept 2024 – Expected Aug 2028

Amrita Vishwa Vidyapeetham

- B. Tech in Computer Science and Engineering (AI)

Chennai, India

Oct, 2020 - May, 2024

CGPA: 9.41 / 10 | Gold Medal, Final Year | Undergraduate Thesis: Cancer Diagnosis and Biomarker Prediction Using Quantum Machine Learning

AWARDS & SCHOLARSHIPS

- **NSERC Canada Graduate Research Scholarship – Doctoral (CGRS-D), 2026** - National doctoral research scholarship, \$40,000/year.
- **DAAD International Research Fellowship, 2026** - Selected as one of 12 fellows from Canada for research at Forschungszentrum Jülich, Germany.
- **Ontario Graduate Scholarship (OGS), 2025–2026** - \$15,000.
- **Vector Scholarship in Artificial Intelligence, 2024–2025** - \$17,500.
- **DAAD RISE Professional Scholar, 2026** - Competitive international research internship in Germany.
- **3MT Winner, 2026** - Presented AI/genomics research to a non-specialist audience.
- **Erin Angus Graduate Student Engagement Award, 2026** - Recognized for leadership and community impact.
- **Project SOY Plus, 3rd Place, 2026** - Built an AI-based pollinator detection system.

EXPERIENCE

Visiting AI Research Fellow - DAAD Fellow

June 2026 - Sept 2026

Forschungszentrum Jülich (FZJ), Institute of Neuroscience and Medicine (INM-4), Germany

- Conducted interdisciplinary research in machine learning, computational neuroscience, and multimodal biomedical data analysis.
- Developed **Python-based machine learning pipelines** for scientific and biomedical data, with a focus on interpretability, reproducibility, and scalable analysis.
- Collaborated with international researchers on multimodal AI methods for complex biomedical and scientific workflows.
- Investigated **graph neural networks, foundation models, and multimodal learning approaches** for next-generation scientific AI applications.

Graduate Research Assistant - Machine Learning & Computational Biology

University of Guelph

Sept 2024 - Present

- Developing **transformer-based foundation models in Python/PyTorch** for biological sequence understanding and antimicrobial peptide discovery.
- Designing interpretable **representation-learning and deep-learning methods** for genomic classification across diverse organisms.
- Building reproducible ML pipelines integrating large-scale biological sequence datasets, metadata, and scientific knowledge.
- Developing and benchmarking **Transformers, CNNs, BiLSTMs, and GRUs** for biological sequence modelling and classification.
- Investigating **LLM reasoning and representation behaviour through hidden-state analysis**, with a focus on interpretability and internal model dynamics.

Software Engineer - Centre for Biodiversity Genomics (CBG)

Jan 2025 - Present

- Designed and implemented a **Python/FastAPI microservice** for DOI generation, metadata validation, and automated processing across large object repositories.
- **Improved data processing efficiency by ~40-60%** through batch metadata extraction, multi-file handling, and resumable uploads/downloads.
- Developed **Python-based bioinformatics tools** for genomic trace-file processing (AB1/SCF) and metadata modelling to support large-scale genomic analysis and downstream ML research.
- Built reproducible software workflows using **Python, REST APIs, Git, Linux, and Docker**.

AI Scientist-in-Residence - NEXT Canada

Mar 2025 - Present

- Advised 30+ startups across AI, healthcare, biotechnology, legal technology, sustainability, and data platforms.
- Advised AI and technology startups on **machine-learning architectures, feature engineering, generative AI applications, model evaluation, and inference optimization**.

Graduate Teaching Assistant

Sept 2024 - Present

University of Guelph

- Conducted tutorials and laboratory sessions, held office hours, and evaluated assignments, projects, and examinations.
- Mentored students in programming, software engineering, and artificial intelligence concepts.

Software Developer Intern - Microsoft

May 2023 - July 2023

Telangana, India

- Migrated the Email Notification Service for the Service Cloud from **.NET 4.8 to .NET 6.0**.
- Improved system scalability to support $\sim 2\times$ higher concurrent workloads, reduced development effort by $\sim 30\%$ through modernization, enabled cross-platform deployment (Windows/Linux), and introduced containerization using Docker for consistent environment setup.
- **Technologies:** C#, .NET 6.0, WebJob SDK, Kusto Query Language, Azure Cloud, Docker

Research Intern - Mitacs

July 2023 - Oct 2023

Prince Edward Island, Canada

- Applied machine learning and **ML-DSP alignment-free sequence analysis** to genomic datasets for identification of cancer-associated biomarkers.
- Developed computational workflows for genomic feature extraction, classification, and comparative analysis.

SELECTED PUBLICATIONS & RESEARCH

- **Measuring Deep Thought in LLMs Through Hidden State Analysis** - Spandana Cherreddy, Nicholas Vinden, Kayla Kopel, Kamal Aslam, Gurjit S. Randhawa, Paul Sheridan, Aitazaz A. Farooque - **ICPRAI 2026**.
- **ALERT: ADML - Antimicrobial Learning Exploration & Research for Targets: Antimicrobial Peptide Discovery with Machine Learning and Deep Learning** - Spandana Cherreddy et al. **Under Review, Computers in Biology and Medicine, Elsevier, 2026**.
- **Integrating Quantum Representation Learning with Graph Intelligence for Molecular Property Prediction in Neurodegenerative Drug Discovery** - Shivanya Shomir Dutta, Spandana Cherreddy, Sridevi S., Gurjit S. Randhawa, Anandan P. **Manuscript in Preparation, 2026**.
- **AI-Driven Analysis of Brain Metabolite Networks in Major Depressive Disorder** - Spandana Cherreddy, Ravichandran Rajkumar, Gurjit S. Randhawa, et al. **Manuscript in Preparation, 2026**.
- **Graph-Based Multimodal Metabolomics for Precision Psychiatry** - Spandana Cherreddy, Ravichandran Rajkumar, Gurjit S. Randhawa, et al. **Manuscript in Preparation, 2026**.
- **Graph Neural Networks for Metabolite Network Learning in Neuropsychiatric Disorders** - Spandana Cherreddy, Ravichandran Rajkumar, Gurjit S. Randhawa, et al. - **Manuscript in Preparation, 2026**.
- **Tweeting the Blues: Leveraging NLP and Classification Models for Depression Detection** - Spandana Cherreddy, K. Geetha, A. G. Sreedevi - **2nd International Conference on Advancement in Computation & Computer Technologies, IEEE, 2024**.
- **Enhancing Computation Offloading in Wireless-Powered Mobile-Edge Computing Networks with Deep Reinforcement Learning for Online Optimization** - R. P. Kumar, Spandana Cherreddy, I. V. Srisurya, A. R. Priyadharshini, G. D. Sai - **Fourth International Conference on Advances in Electrical, Computing, Communication and Sustainable Technologies, 2024**.
- **Retroactive Data Structure for Protein–Protein Interaction in Lung Cancer Using Dijkstra Algorithm** - P. K. Rangarajan, B. M. Gurusamy, E. Rajasekar, I. V. Srisurya, Spandana Cherreddy et al. **International Journal of Information Technology, 16(2), 1239–1251, 2024**.
- **An Efficient Genetic Algorithm-Based AutoML Approach for Classification and Regression** - Spandana Cherreddy, I. V. Srisurya, S. A. Nandhini, R. P. Kumar, G. B. Mohan et al. **International Conference on Intelligent Data Communication Technologies and Internet of Things, 2023**.
- **Underwater Image Enhancement and Restoration Using CycleGAN** - Spandana Cherreddy, I. V. Srisurya, A. R. Priyadharshini, S. Krithika, S. A. Nandhini et al. - **International Conference on Innovative Computing and Communications, Springer, 2023**.
- **Detection of Cancer in Human Blood Sample Using Machine Learning** - Spandana Cherreddy, R. P. Kumar - **7th International Conference on Computing Methodologies and Communication (ICCMC), IEEE, 2023**.
- **Application of Machine Learning and Deep Learning Algorithms in Predicting Virtual Network Functions for Network Function Virtualization** - Spandana Cherreddy, I. V. Srisurya, S. Sridevi - **14th International Conference on Computing Communication and Networking Technologies, IEEE, 2023**.

- **Predicting the Driver Variants and Mutations in Lung Cancer Genome Using Transcriptional Regulation Network** - Spandana Chereddy, I. V. Srisurya, H. Bogineni - **7th International Conference on Computing Methodologies and Communication, IEEE, 2023.**
- **A Variational Autoencoder–Generative Adversarial Networks (VAE-GAN) Based Model for Ligand Designing** - K. Mukesh, I. V. Srisurya, Spandana Chereddy, E. Anbazhagan, I. R. Oviya - **International Conference on Innovative Computing and Communications, Springer, 2022.**
- **Chest X-ray Pathology Detection Using Deep Learning and Transfer Learning** - I. R. Oviya, Spandana Chereddy - **7th International Conference on Recent Advances and Innovations in Engineering (ICRAIE), IEEE, 2022.**

Full publication list and citation metrics available on Google Scholar.

- **WiML @ ICML 2026** - Poster presentation on scalable and interpretable machine learning for genomic sequence classification.
- **ICEM 2026, Edinburgh** - Poster presentation on machine learning for genomic sequence classification / computational biology.
- **CBHC 2026, Toronto** - Poster presentation on scalable ML methods for biological sequence analysis.

LEADERSHIP AND SERVICES

Vice President | Graduate Student Association (GSA), University of Guelph *May 2025 – July 2026*

- Represent over 3,000 graduate students and collaborate with university leadership on governance, policy, student engagement, and organizational initiatives.

Member | Information Technology Student Advisory Council (ITSAC), University of Guelph

Sept 2024 – Present

- Serve as an advisory member, providing input on IT policies, strategies, and initiatives that affect students across the university.

Canadian Union of Public Employees (CUPE 3913) – Steward, School of Computer Science

Jan 2025 – April 2025

- Representing graduate student employees, advocating for workplace rights, and facilitating communication between CUPE members and university administration.

SKILLS

- **Machine Learning & AI:** Machine Learning, Deep Learning, Transformers, Foundation Models, Generative AI, Representation Learning, Multimodal Learning, Explainable AI, CNNs, RNNs
- **Programming Languages:** Python, C/C++, Java, C#, SQL
- **Frameworks & Libraries:** PyTorch, TensorFlow, scikit-learn, NumPy, Pandas
- **Research & Data:** Scientific AI, Computational Biology, Genomics, Bioinformatics, Data Science, Biological Sequence Analysis
- **Software & Systems:** FastAPI, REST APIs, Docker, Linux, Git, AWS, Azure